

Title

Association between parental oral health beliefs and cavity disease occurrence in elementary school children

Abstract

Background: Parental factors influence children's oral health behaviors and disease risk, but associations between specific parental psychosocial constructs and pediatric caries remain incompletely characterized. **Methods:** This cross-sectional analysis examined baseline data from 1084 parent-child dyads (34 elementary schools, Texas). Dental exams used standardized caries detection criteria. Parental knowledge, self-efficacy, and child oral health-related quality of life were assessed via validated surveys. **Results:** Higher parental self-efficacy was independently associated with lower caries prevalence in children (adjusted odds ratio 0.82, 95% CI 0.74-0.91, $p=0.02$). Parental oral health knowledge showed no significant independent association with child caries prevalence after adjustment. Child oral health-related quality-of-life scores correlated significantly with caries status ($p<0.001$). Demographic factors including family income and maternal education influenced both parental self-efficacy and child caries risk. Girls showed lower caries prevalence than boys (21.8% vs 28.9%, $p=0.004$). **Conclusions:** Parental self-efficacy regarding child oral hygiene emerged as a protective factor against early childhood caries, suggesting interventions targeting parental confidence in preventive capacity may reduce disease burden.